Project Proposal for Package R: 'BFRdist'

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1. Description and Context

This package aims to capture the functionality of the BFR distribution. The BFR distribution is a novel, two-parameter model which is arctan-generated and defined on the unit interval for double bounded data (Bhattacharya et al., 2026). Specifically, this project implements the d/p/q/r family of functions for the BFR distribution and generates the novel Pearson-like diagnostic plot (based on Galton's skewness and Moors' kurtosis) proposed by Bhattacharya et al. (2026) alongside the distribution. Plotting functions will be implemented using base R graphics, ensuring the package remains lightweight with minimal external dependencies. Ultimately, this package translates a novel theoretical model into a practical, ready-to-use computational tool, allowing researchers to easily apply the BFR distribution to their own data.

User Inputs and Outputs

Users of the `BFRdist` package will primarily input clean, numeric vectors containing values strictly greater than 0 and less than 1 to fit the BFR distribution to their observations. By interacting with the package functions, users can expect to output estimated parameters (α and β), calculate summary statistics, and generate visualizations comparing empirical data to the fitted theoretical distribution.

2. Mathematical Formulation (Bhattacharya et al., 2026)

The BFR distribution's density, distribution, quantile, and random generation functions can be defined in the following formulas :

Probability Density Function (PDF)

$$f_{BFR}(x;\alpha,\beta) = \frac{2\beta}{\pi} \cdot \frac{\alpha^{\beta} x^{\beta-1}(1-x)^{\beta-1}}{\alpha^{2\beta}x^{2\beta} + (1-x)^{2\beta}} 1_{(0,1)}(x),$$

where $\alpha > 0$ and $\beta > 0$.

Cumulative Distribution Function (CDF)

$$F_{BFR}(x;\alpha,\beta) = \begin{cases} 0 & \text{if } x \leq 0 \\ \frac{2}{\pi}\arctan\left(\frac{\alpha x}{1-x}\right)^{\beta} & \text{if } 0 < x < 1 \\ 1 & \text{if } x \geq 1 \end{cases},$$

where $\alpha > 0$ and $\beta > 0$.

Quantile Function

$$Q_{BFR}(u;\alpha,\beta) = \frac{1}{1 + \alpha\left(\tan\left(\frac{\pi u}{2}\right)\right)^{-1/\beta}} \mathbb{I}_{(0,1)}(u),$$

where $\alpha > 0$ and $\beta > 0$.

Random Generation Function

First, generate a realization u from $U \sim \text{Uniform}(0, 1)$.

Then compute:

$$X = Q_{\text{BFR}}(u; \alpha, \beta) = \left[1 + \alpha \left(\tan\left(\frac{\pi u}{2}\right)\right)^{\frac{-1}{\beta}}\right]^{-1},$$

and therefore, $X \sim \text{BFR}(\alpha, \beta)$.

3. Package Deliverables and Architecture

Deliverables

1. **Core functions:** Implementation of `dbfr`, `pbfr`, `qbfr`, and `rbfr`.
2. **Summary Statistic functions:** Galton Skewness and Moors Kurtosis.
3. **Base Estimation functions:** Parameter inference using Maximum Likelihood Estimation (MLE) and Quantile Estimation (QE) via `fit_bfr`.
4. **C++ Quantile-based Estimators:** High-performance parameter inference using numerical root-finding algorithms.
5. **Visualisation Function:** Pearson-like SKG/KUM diagrams.
6. **Unit Tests & Documentation:** Verify package validity with testing on all core and estimation functions, and feature a comprehensive vignette to guide the user.

Object-Oriented (S3) Implementation

To leverage the object-oriented features of R, our estimation functions will return a custom object of class `bfr`. We will implement four object-oriented S3 generic methods for this class:

- `print.bfr`: To display basic fitted parameter estimates.
- `summary.bfr`: To output standard errors and goodness-of-fit metrics.
- `plot.bfr`: To generate Pearson-like plot(s) automatically.
- `predict.bfr`: To simulate new predictions based on the fitted model.

Compiled Code (C++) Advantages

Computationally expensive processes will be outsourced to C++ functions inside the `src` folder via Rcpp for the quantile-based estimation. By using Rcpp, we bypass R's slower base loops, gaining substantial execution speed and memory efficiency.

Software Diagrams

Figure 1 illustrates the foundational folder structure of the package, highlighting the standard CRAN-compliant organization of R scripts, compiled C++ source files (`src/`), and testing directories. Building upon this architecture, Figure 2 maps the sequential workflow of the package deliverables, demonstrating how the core mathematical functions and summary statistics feed directly into the estimation, simulation, and visualization pipelines.



Figure 1: Package Folder Structure

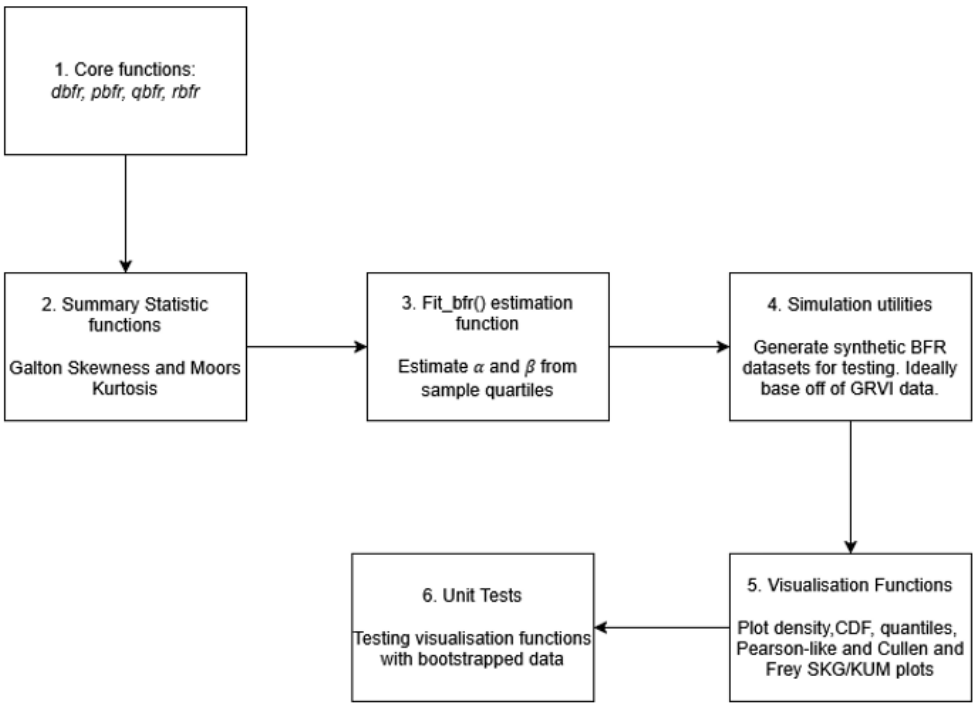


Figure 2: Package Deliverables

Timeline and Work Distribution

To ensure a structured and timely development process, Table 1 delineates the project schedule and individual responsibilities. The table tracks milestones from the initial implementation of core functions in Week 6 through the finalization of unit tests and documentation in Week 12, establishing clear deadlines and balanced contributions between both developers.

Table 1: Projected Timeline and Work Distribution

Deliverable	Primary Contributor	Target Completion
1. Core functions (d/p/q/r)	Jennifer Hanna	Week 6
2. Summary Statistics	Jennifer Hanna	Week 7
3. Base Estimators (ML & Moments)	Jennifer Hanna	Week 8
4. Quantile-based Estimators (C++)	Daniel Luo	Week 10
5. Visualisation Function	Daniel Luo	Week 11
6. Unit Tests & Documentation	Daniel Luo	Week 12

A working draft package will be submitted in Week 10 featuring core functions, draft documentation (vignette), and a detailed test plan for peer review. The final package (Week 13/14) will deliver the fully refined suite of all 6 project deliverables, including finalized unit testing.

References

- Bhattacharya, A., Frery, A. C., & Rey, A. A. (2026). A Statistical Framework for Bounded Measurements with Applications to SAR Data. *IEEE Transactions on Geoscience and Remote Sensing*.